RAPIVOLT

ON-DEMAND EV CHARGING PLATFORM

Project Concept, Rationale and Implementation Approach

PROJECT DOCUMENTATION

Team: The Chaos

Team Member	Roll Number
Suraj Kumar	2402310100098
Rohit	2402310100080
Priya Sharma	2402310100076
Vaishnavi	2402310100112

RapiVolt – Making EV Charging Accessible, Anywhere.

PROJECT OVERVIEW

RapiVolt is a proposed on-demand EV charging platform that connects electric-vehicle drivers with nearby registered homeowners and professional charging providers. The platform is designed to support location-based discovery, charger compatibility checks, availability and price comparison, booking requests, and charging-session records.

The initial release is planned as a single web application with two primary roles: Driver and Charging Provider. Some capabilities—such as live charger integration, online payment processing, automated notifications and smart recommendations—are identified as future enhancements or dependent on external integrations.

Document at a glance

- WHAT: The platform, its users and its core service.
- WHY: The user needs and opportunity the project addresses.
- HOW: The end-to-end workflow, functional modules and proposed implementation.
- CHALLENGES: Seventeen foreseeable technical, operational, safety and trust-related challenges with proposed responses.
- SCOPE: Features planned for the initial release and possible future enhancements.

1. WHAT – What is RapiVolt?

RapiVolt is an on-demand Electric Vehicle (EV) charging platform that connects EV drivers with nearby registered homeowners and professional charging providers. It allows drivers to discover charging facilities, review charger compatibility, compare availability and prices, and request charging services according to their location and requirements.

Providers can register charging facilities, add charger information, manage operating hours and availability, set prices, and accept or reject incoming requests. Initially, RapiVolt will operate as a single web application with two primary user roles:

- Driver: A user who needs a charging service for an electric vehicle.
- Charging Provider: A homeowner or professional charging business that offers a charging facility.

A single user account may support both roles, subject to the application's role and profile setup. Each provider may manage one or more facilities, and each facility may contain multiple individually managed chargers.

The objective is to create a convenient and structured connection between people who need EV charging and those who can provide it.

2. WHY – Why are we building RapiVolt?

As electric-vehicle use grows, finding convenient and accessible charging facilities can remain a challenge. Drivers may struggle to locate a nearby charger, particularly when their vehicle battery is low or public charging stations are unavailable or far away. At the same time, homeowners and private charging providers with suitable facilities may have an opportunity to earn additional income by offering charging services.

RapiVolt aims to address these needs by:

- Making nearby charging facilities easier to discover.
- Connecting drivers with providers who have available charging capacity.
- Helping drivers review vehicle–charger compatibility before booking.
- Showing the applicable tariff and an estimated cost before a booking is confirmed.
- Reducing dependence on traditional public charging stations where suitable alternatives exist.
- Creating earning opportunities for homeowners and professional charging providers.

- Supporting the wider adoption of electric mobility.

3. HOW – How will RapiVolt work?

RapiVolt will operate through a location-based discovery and booking system. It will compare the driver's vehicle requirements with provider-submitted charger details, availability and pricing. The following workflow describes the proposed process:

Step 1: User registration and login. Users create an account and log in. They may use the application as a Driver, a Charging Provider, or both if their account is configured for both roles.

Step 2: Provider registration. A provider submits facility location, charger type, connector details, power rating, pricing, operating hours and availability. Provider and facility details remain subject to verification.

Step 3: Vehicle registration. Drivers add relevant EV information, including vehicle model and charging-related compatibility details.

Step 4: Finding nearby providers. Drivers share their location or enter a search location. RapiVolt displays relevant facilities with distance, charger details, compatibility information, prices and availability.

Step 5: Compatibility and selection. Drivers review the options and select a facility that meets their vehicle requirements, location, availability and price preferences.

Step 6: Booking request. The driver selects an available time slot, reviews the tariff and estimated cost, and sends a charging request.

Step 7: Provider response. The provider receives the request and can accept or reject it based on availability and facility conditions. The booking status is updated accordingly.

Step 8: Booking confirmation. Once accepted, the booking is confirmed. The driver receives booking details and relevant access instructions. Exact residential address details are shared only after confirmation, subject to the privacy policy and access controls.

Step 9: Charging session. The driver receives the charging service. Session status and relevant details are recorded. Where compatible metering or equipment is available, electricity consumption can also be recorded.

Step 10: Completion and payment record. After the session, booking and charging statuses are updated. Where valid meter readings are available, the final cost can be calculated using the recorded energy consumption and agreed tariff. The application records the payment or transaction status supported by the release.

Proposed booking status flow

A booking may move through the following statuses, depending on the action taken:

- Requested – the driver has submitted a booking request.
- Accepted / Confirmed – the provider has accepted the request.
- Rejected – the provider has declined the request.
- Cancelled – the booking has been cancelled according to the applicable policy.
- Completed – the charging service has been marked complete.
- Interrupted – the charging session has stopped before normal completion, where session tracking is supported.

The application should maintain a status history so that important booking changes can be reviewed.

4. CHARGING EXPERIENCE

A. Wait & Charge

Drivers remain at or near the charging facility while their vehicles are being charged. The platform may show estimated charging duration and cost. It may also help drivers discover nearby cafes, restaurants, shopping

centres, stores and places of interest. Completion notifications and updated estimates depend on suitable system or hardware integration.

B. Drop & Charge

Drivers leave their vehicles with eligible and verified charging providers and return after the session is completed. This option requires secure parking, documented vehicle-condition records, handover and collection confirmation, and clearly established liability and insurance arrangements. It should only be offered by providers who meet the platform's eligibility requirements. These operational and legal requirements must be defined before this service is enabled.

5. PRIVACY, SAFETY AND TRUST

Because RapiVolt may connect drivers with residential and commercial charging providers, privacy and safety must be considered throughout the booking process. Proposed safeguards include:

- Provider-controlled availability: Providers manage their operating hours, available slots and access conditions.
- Optional night charging: Providers decide whether to offer night-time charging.
- Provider and facility verification: Identity and facility details are reviewed through an authorized verification process.
- Booking approval: Providers retain control over accepting or rejecting requests.
- Location privacy: Exact residential addresses are disclosed only after booking confirmation and only to authorized users. This restriction must be enforced by backend access controls, not merely by hiding information on the screen.
- Safer facility access: Separate outdoor parking or charging spaces are encouraged where feasible.
- Emergency support: Users can report safety concerns through a defined support and incident-reporting process.
- Drop & Charge safeguards: Secure parking, vehicle-condition records, handover and collection confirmation, and documented liability and insurance requirements are necessary before enabling this mode.
- Ratings and reviews: These are planned as a later-release feature, as described in the Future Scope section.

Proposed provider verification workflow

1. The provider submits identity and facility information with any required supporting details.
2. An authorized platform administrator reviews the submitted information.
3. The provider or facility is assigned a status: Pending, Approved or Rejected. A Suspended status may be used if an already approved facility must temporarily stop offering service.
4. Only approved providers and facilities are made bookable.
5. Reported safety concerns may trigger a review and temporary suspension until the issue is checked.

6. FUNCTIONAL MODULES

Module	Responsibility
User Management	Registration, login, profiles and role-based access.
Vehicle Management	Recording EV model and charging-related information.
Provider Management	Provider profiles, submitted facility details and verification status.
Facility and Charger Management	Managing facilities and individual chargers, connector types, power details and charger status.
Location and Search	Nearby facility discovery and distance display.

Compatibility	Comparing vehicle requirements with charger information before booking.
Availability and Booking	Time slots, requests, acceptance, rejection, cancellations and booking status history.
Pricing	Displaying the applicable tariff and preserving the agreed price for a confirmed booking.
Charging Session	Session start, end, status and energy readings where suitable metering is available.
Safety and Support	Address privacy, verification, incident reporting and Drop & Charge handover records.
Payment Records	Recording basic payment or transaction status in the initial release; online payment gateway and automated settlement integration are future enhancements.

7. CHARGING COST CALCULATION

Where compatible energy-meter readings are available, energy consumed can be calculated from the difference between the ending and starting meter readings.

Energy Used (kWh) = Ending Meter Reading − Starting Meter Reading

Charging Cost = Energy Used (kWh) × Agreed Price per kWh

Example: Starting reading = 120 kWh; ending reading = 130 kWh; energy consumed = 10 kWh; agreed tariff = ₹15/kWh. Calculated energy charge = 10 × ₹15 = ₹150.

The displayed amount before charging should be treated as an estimate unless the actual energy consumed is already known. The final energy-based charge depends on valid readings and the agreed tariff. Any additional fee, if applicable, must be disclosed before booking and handled according to the agreed policy. The confirmed booking should retain its agreed tariff so that a later change to the provider's public tariff does not alter that booking.

This example illustrates the calculation only; it does not define taxes, platform fees, parking fees or other charges.

8. PROPOSED TECHNOLOGY APPROACH

The initial version is proposed as a single web application. The following stack is the current proposed approach and should be treated as provisional until confirmed by the project team:

Layer	Proposed technology
Frontend	HTML, CSS, JavaScript and Bootstrap
Backend	Java Servlets and JSP
Database	MySQL
Database connectivity	JDBC
Application server	Apache Tomcat
Development environment	Eclipse IDE

At a high level, the browser-based interface sends user actions to the Java application layer. The application layer validates input, applies booking and access rules, and uses JDBC to communicate with the MySQL database. Apache Tomcat hosts the web application. The final architecture, tools and versions should be confirmed by the team before implementation.

External services—such as maps/geolocation, payment gateways, email/SMS notifications and charger telemetry—will require separate integration decisions and may be introduced in later releases.

9. POTENTIAL CHALLENGES AND PROPOSED SOLUTIONS

The following points describe foreseeable operational and technical challenges and proposed measures. Some measures require provider verification, operational policies, qualified physical inspection or compatible external equipment in addition to application development.

1. Privacy and security concerns

Problem: Homeowners may hesitate to allow unknown people onto residential property, particularly at night, because of personal safety and privacy concerns.

Proposed approach:

- Verify user identities and provider details through an appropriate process.
- Allow providers to control operating hours and access permissions.
- Prefer separate outdoor charging spaces where feasible.
- Share exact residential addresses only after booking confirmation and enforce this restriction through backend authorization.
- Provide incident reporting and a defined support process.

2. Charging compatibility issues

Problem: Different EVs support different connectors and power requirements, so a listed charger may not suit a particular vehicle.

Proposed approach:

- Collect relevant vehicle details during driver registration.
- Collect charger type, connector and power details during provider registration.
- Check compatibility before allowing a booking request.
- Show the basis of the compatibility result and avoid presenting unverified compatibility as confirmed.

3. Provider unavailability after booking

Problem: A provider may accept a request but later become unavailable when the driver arrives.

Proposed approach:

- Confirm availability for a defined time slot.
- Associate each confirmed booking with a specific charger or confirmed charging capacity.
- Use booking and arrival confirmation where supported.
- Record provider cancellations and make the booking status visible.
- Show alternative providers where available.

4. Unexpected pricing and additional charges

Problem: A provider may request extra charges after booking, causing disagreement.

Proposed approach:

- Display the applicable tariff and estimated total before booking.
- Clearly state whether pricing is per kWh, per session or includes another disclosed fee.
- Preserve the agreed tariff and disclosed charges for the confirmed booking.
- Require driver approval for any permitted additional charge before it is applied.
- Maintain transaction records and receipts where supported.

5. Electricity consumption disputes

Problem: Providers may be concerned about electricity bills, while drivers may question the amount of energy recorded.

Proposed approach:

- Use compatible energy meters or smart charging equipment where available.
- Record kWh consumption only when reliable readings are available.
- Maintain session start and end readings with timestamps where supported.
- Show the calculation used to determine the energy charge.
- Provide a process to flag disputed or missing readings rather than treating estimates as measured consumption.

6. Booking cancellations and no-shows

Problem: Drivers may not arrive after booking, or providers may cancel at short notice.

Proposed approach:

- Define a clear cancellation and no-show policy before booking.
- Send reminders where notification capability is available.
- Maintain cancellation records and the reason where provided.
- Apply only fair, clearly disclosed charges where appropriate.
- Use proportionate restrictions for repeated misuse.

7. Vehicle theft or damage during Drop & Charge

Problem: Drivers may hesitate to leave vehicles with an unfamiliar provider.

Proposed approach:

- Allow Drop & Charge only through eligible, verified providers.
- Record vehicle condition at handover, with driver and provider acknowledgement.
- Maintain handover and collection confirmation.
- Require secure parking and defined vehicle-access procedures.
- Establish liability and insurance requirements before offering this service.

8. Unexpected charging-duration differences

Problem: Actual charging duration may differ from the initial estimate because of battery condition, charger power, temperature or other factors.

Proposed approach:

- Calculate estimates using available vehicle and charger information, while stating the assumptions.
- Present duration as an estimate, not a guaranteed completion time.
- Update estimates only when suitable live session data is available.
- Notify drivers about known delays when the system receives reliable updates.

9. Lack of nearby available providers

Problem: Drivers may not find a provider in their immediate area.

Proposed approach:

- Allow the driver to expand the search radius progressively.
- Include approved commercial providers in the search results.
- Suggest other available facilities where possible.
- Support scheduled bookings where providers offer them.

- Encourage provider registration in underserved areas.

10. Electrical safety and equipment quality

Problem: Improper wiring, unsuitable equipment or overloaded connections may create electrical hazards.

Proposed approach:

- Require an appropriate facility verification process before approval.
- Require suitable charging equipment for the declared use.
- Make physical inspection by qualified electrical professionals a prerequisite where applicable.
- Have qualified professionals check earthing and electrical protection.
- Follow applicable standards and local requirements.
- Temporarily suspend facilities reported as unsafe until they have been checked.

11. Charger already occupied

Problem: An individual charger may already be reserved, in use or out of service even when the provider is otherwise active.

Proposed approach:

- Track status separately for each charger, not only at provider level.
- Use states such as Available, Reserved, Charging and Out of Service.
- Associate bookings with a specific charger or confirmed capacity.
- Prevent conflicting bookings for the same charger and time slot.
- Update charger status when a session starts, ends or is interrupted.

12. Driver cannot reach the charging location

Problem: A selected facility may be too far away or difficult to reach with the vehicle's remaining range.

Proposed approach:

- Show the facility location and distance before booking.
- Offer closer alternatives where available.
- Show travel-energy or remaining-range estimates only when reliable vehicle and route data are available.
- Consider emergency mobile charging as a future service, subject to feasibility and safety requirements.

13. Electricity failure during charging

Problem: A power interruption may stop or delay a charging session.

Proposed approach:

- Record the interruption and update the session status when the system is informed.
- Notify the driver and provider when the interruption is known to the system.
- Calculate energy charges from valid recorded consumption rather than assuming the planned amount was delivered.
- Show alternative facilities where available.

14. Driver does not collect the vehicle

Problem: A driver may not return promptly after a Drop & Charge session is completed.

Proposed approach:

- Send a completion notification where supported.
- Define a collection deadline and reasonable grace period.
- Disclose any late parking or storage charges before booking.
- Provide a support route for collection issues.
- Record collection confirmation.

15. Fake charging providers

Problem: Fraudulent users may list facilities they do not own or cannot provide.

Proposed approach:

- Verify provider identity and facility details before approval.
- Request supporting information where appropriate.
- Require an approved status before a facility becomes bookable.
- Allow users to report suspicious providers.
- Review reports and suspend listings where necessary while concerns are investigated.

16. Payment failure and provider settlement delay

Problem: A transaction may fail, remain pending or require a refund; provider settlement may also be delayed.

Proposed approach:

- Assign a unique reference to each payment or transaction record.
- Track statuses such as Pending, Successful, Failed and Refunded where applicable.
- Maintain transaction and refund records.
- Keep provider settlement records separate from driver payment records.
- Treat full payment-gateway processing and automated provider settlement as future scope unless included in the initial release.

17. Charging session exceeds planned duration

Problem: A session may continue beyond its planned end time, affecting the driver's schedule and the next booking for that charger.

Proposed approach:

- Track the session's actual status and recorded end time where supported.
- Update charger availability only after the session status is appropriately changed.
- Notify affected users about known overruns where notification capability exists.
- Check for conflicts with the next booking and provide an alternative or revised slot where available.
- Keep this operational issue distinct from the accuracy of the original charging-duration estimate.

10. INITIAL IMPLEMENTATION SCOPE

The first release should focus on the essential workflow required to discover a provider, request and confirm a booking, receive charging service and record the session.

- Single web application with Driver and Charging Provider roles.
- Registration, login, profiles and role-based access.
- Vehicle and provider/facility registration.

- Provider verification status and individual charger information.
- Nearby provider search and compatibility display based on recorded details.
- Provider availability, booking requests and accept/reject actions.
- Booking confirmation, status history and controlled sharing of residential address details.
- Agreed pricing and estimated cost display.
- Charging-session records and energy-consumption records where suitable readings are available.
- Basic payment or transaction record keeping, without implying that an online payment gateway is already integrated.
- Essential privacy, safety and incident-reporting procedures.

The exact features included in the first release should be finalized by the project team after reviewing available time, hardware access and integration requirements.

11. FUTURE SCOPE

As RapiVolt develops, the platform may be expanded with the following capabilities:

- Separate applications or tailored interfaces for Drivers and Charging Providers.
- Live charging progress, battery percentage and estimated completion time where charger or vehicle integration supports it.
- Smart recommendations based on distance, price, availability and charger compatibility.
- Secure online payment gateway integration, digital receipts and automated provider settlement.
- Advanced real-time booking and charging notifications.
- Ratings and reviews.
- Detailed charging, booking and transaction history.
- Emergency mobile charging assistance, subject to feasibility and safety requirements.
- Smart energy monitoring and detailed consumption information.
- Nearby cafes, restaurants, shopping areas and other useful places as a charging-time companion.
- Expansion to more cities and charging networks.

12. OUR VISION

Our vision is to make EV charging more accessible, convenient and reliable by creating a connected charging ecosystem. RapiVolt aims to simplify the charging experience for EV drivers while providing homeowners and professional charging providers with new earning opportunities.

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